

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

6 September 2026

Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Declared actuator modes bound mean coupling by installed energy, giving a necessary resource condition for the scalar synchronization threshold. Phase encoders shared on overlaps convert coherence into a vector-content disagreement bound. Compatible spatial measures glue into one state; a contracting reconstruction map that fixes it makes it the unique fixed point. Reconstruction across a declared state family additionally requires distinguishable encodings, and local communication limits timely responses to interventions. Lean 4 checks these results and two compositions, each with eight connecting hypotheses and additional structures. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. The awakening hypothesis links extracellular geometry to increasing coupling. Finite ramps delay stationary onset, while coupling increase, diffusion decrease and a matched common drive can share stationary phase distributions. Testing requires calibrated physical identifications, resolved

dynamics and independent measurements of content agreement and self-encoding.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Consider reaching for a cup. A visual description of its location and a description of the hand's relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation.

21 This example motivates the requirements; the mathematical constructions
22 use spatial measures and fields in real normed vector spaces, whose identifi-
23 cation with perceptual content is a further modelling task.

24 The results specify both connections and separations between these re-
25 quirements. A declared actuator’s stored energy bounds its mean coupling,
26 giving a necessary energy condition for the scalar coherence threshold. Equal
27 phases permit incompatible local descriptions when content is independent
28 of phase; when a common Lipschitz encoder reads phase, coherence bounds
29 disagreement. Exact compatibility yields a unique global extension, and ap-
30 proximate compatibility bounds the variation among specified selections. A
31 contracting reconstruction map has a unique fixed point, with a quantita-
32 tive error bound, but identifying that point with the state formed from local
33 descriptions requires accuracy at that state.

34 Awakening gives the physical hypothesis a concrete test. We propose that
35 changing extracellular geometry increases coupling through the synchroniza-
36 tion threshold. The stationary onset is square-root, but finite ramp speeds
37 delay recovery. Moreover, increasing coupling, decreasing diffusion and a
38 matched common drive can produce identical stationary phase distributions.
39 A separate plasticity control preserves coherence while losing alignment with
40 its diagnostic environmental template. These results identify measurements
41 and mechanisms that a successful coordination model must supply.

42 The article follows the three requirements and their tests (Fig. 1). Ther-
43 modynamic constraints specify the costs of candidate implementations, in-

cluding agents that act on their future inputs. The detailed feedback, learning and resource models have distinct state spaces and assumptions; their consequences do not together instantiate a cortical agent. Section 6 summarizes the conditional composition, and Supplemental Material provides the constructions, witnesses and numerical controls.

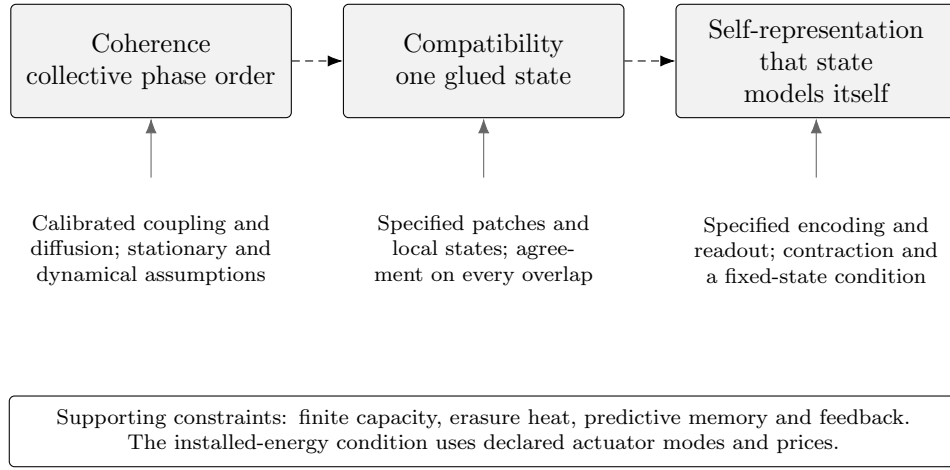


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

49 2 Coherence: physical coupling and its re- 50 source condition

51 2.1 Dynamics and the stationary threshold

52 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

53 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
54 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
55 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
56 parameters. A spatially decaying kernel enters the scalar model through its
57 row sum at zero diagonal, and a finite sheet measures what that reduction
58 costs: over decay lengths from 0.1 to 0.3 mm, spanning 87 to 566 effective
59 neighbours, the measured threshold lies at 0.750–0.753 times $2D$, within
60 0.003 of the 0.750 a uniform control returns on the same sheet and fit window.
61 Both sit below $2D$ there — the fit window alone brackets them down to
62 0.658 — so the agreement is between the reduction and its control rather
63 than a confirmation of the threshold’s value. What displaces the threshold
64 is frequency heterogeneity rather than spatial reach: a quenched spread of
65 1.5 rad s^{-1} moves it to 1.53 times $2D$, while the same sheet reproduces the
66 stationary branch to 0.0048 over 19 supercritical couplings.

67 For identical frequencies in their rotating frame, the positive normalized
 68 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

69 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 70 mal derivation uses the absence of stationary current for a periodic gradient
 71 drift to obtain this density, then verifies normalization and self-consistency.
 72 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one
 73 positive solution coexists with zero and increases strictly with K . Its onset
 74 is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

75 Figure 2 shows this stationary branch.

76 Linearization about the uniform density gives rate $K/2 - D$ for the first
 77 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 78 exactly above threshold. About the coherent density, the quadratic form
 79 is strictly dissipative transverse to collective rotation, whose rate is zero.
 80 These linear estimates concern the limiting equation. Relating them to finite
 81 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 82 tion; exchange symmetry alone does not establish the needed independence
 83 (Sznitman, 1991).

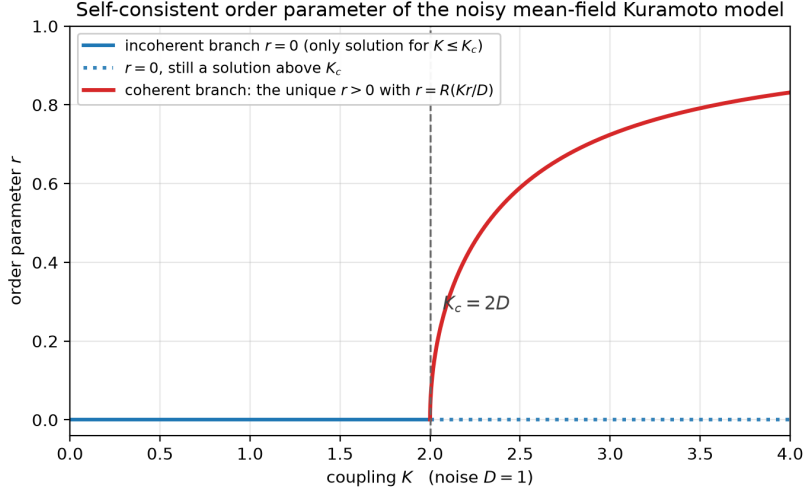


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

84 2.2 Installed energy bounds mean coupling

85 A spatial model replaces the finite sum by integration against a kernel. One
 86 declared implementation carries response modes with continuous profiles ϕ_i ,
 87 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

88 The same configuration sets the reciprocal kernel and stored energy. An
 89 executed transition changes the kernel through its changed occupancies and
 90 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The

91 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 92 pair reconstruction on refining spatial partitions converges to its product-
 93 measure integral under the specified compactness and sampling assumptions;
 94 the refinement index is independent of the number of control updates.

95 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 96 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 97 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

98 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 99 coherent stationary branch and any growing Fourier mode about the uniform
 100 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 101 is necessary: energy above it need not produce sufficient coupling because
 102 the bound can be loose.

103 The quantity constrained is installed energy. Dissipated heat does not
 104 determine it, and κ is supplied by the mode profiles, prices and substrate
 105 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 106 tial dynamics requires a justified reduction. Identifying the installed kernel
 107 with a cortical field is a separate physical commitment.

108 2.3 Physical identification and calibration

109 The cortical hypothesis identifies part of effective coupling with endogenous
110 extracellular electromagnetic interactions. Fields can modulate spike timing
111 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
112 propose coordination of distributed neural information (Pinotsis & Miller,
113 2022; McFadden, 2020). Phase organization under propofol motivates study-
114 ing this regime (Bastos et al., 2021), but does not measure coupling.

115 The formal field predicate requires a jointly continuous kernel, a normal-
116 ized substrate, coupling equal to the kernel average, and no single site pro-
117 viding more than half that strength. Toy constructions satisfy it. A cortical
118 realization must specify the scale at which the kernel and its reduction ap-
119 ply: calcium imaging reports different state-dependent modularity patterns
120 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

121 Finitely many components do not by themselves exclude continuum cou-
122 pling. Spreading a finite coupling matrix over cells of positive measure pre-
123 serves both total coupling weight and phase correlation exactly. Zero cou-
124 pling for finite point support under an atomless measure concerns that repre-
125 sentation of the support. Separately, a wiring constraint forces a correlation
126 disadvantage when it excludes a better-correlated pair. These comparisons
127 supply neither energy prices nor a biological–digital distinction; the supple-
128 ment gives the constructions and controls.

129 A spike-time shift alone supplies no coupling rate. The dimensionless
130 product of population size N , timing sensitivity s , field amplitude E and

131 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (6)$$

132 This rate describes how induced timing shifts accumulate into drift. Nei-
133 ther it nor the installation factor κ follows from a heat bound. Testing the
134 physical threshold requires independent estimates of the interaction kernel,
135 its aggregation and diffusion in the same cortical regime.

136 **3 Testing coherence through recovery after** 137 **awakening**

138 The proposed control variable is extracellular geometry. Sleep-associated
139 changes in extracellular space (Xie et al., 2013) motivate coupling that evolves
140 more slowly than phase dynamics. The cited geometry and phase time scales
141 put that ratio between 600 and 60×10^3 ; at the lower end the stationary
142 reading tracks the branch only down to $|K - K_c| = 0.8$, and at the upper
143 end down to 0.05. Extracellular volume fraction is approximately $\alpha = 0.2$
144 at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the sleep study reports a
145 rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tis-
146 sue conducts approximately resistively at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour, 2017). At
147 fixed transmembrane current density, an effective-medium description pre-
148 dicts that a larger extracellular fraction raises conductivity and reduces field

149 amplitude.

150 The coupling-gated hypothesis consequently predicts that extracellular
151 contraction on awakening raises E and, holding the other factors in Eq. (6)
152 fixed, raises effective coupling. This argument supplies a direction; the
153 geometry–coupling relation, its magnitude and its role in human recovery
154 require measurement.

155 If $K(t)$ crosses K_c while the distribution tracks the coherent stationary
156 branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (7)$$

157 The crossing predicts latency only when coupling is measured independently
158 of coherence. Critical slowing down restricts branch tracking near threshold,
159 and finite populations fluctuate below it. An observable onset therefore needs
160 a criterion above that fluctuation floor.

161 **3.1 What a recovery distribution can identify**

162 Equation (2) gives a concentration–coherence relation with no fitted shape
163 parameter. Estimate r from circular resultant length and a from density
164 shape: inverting R from the measured r would make agreement automatic.
165 Even independently estimated agreement identifies a statistical family shared
166 by different mechanisms.

167 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-

168 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 169 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 170 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 171 and diffusion D_0 have that same density. Their stationary distributions co-
 172 incide window by window despite different causes.

173 Resolved dynamics supply additional information. In the ideal model with
 174 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 175 increment means constrain drift. Separating endogenous coupling from a
 176 matched common drive additionally requires a measured drive or controlled
 177 perturbation; identifying electromagnetic coupling requires evidence specific
 178 to that interaction. Supplemental Material gives the equivalence construction
 179 and its numerical check.

180 **3.2 Finite ramps change the observable onset**

181 The ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling
 182 speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after
 183 threshold. Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape
 184 within the fastest window. The three slower ramps have mean coupling
 185 delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents run from 0.898 to
 186 0.417, compared with 0.443 to 0.448 for the exact stationary branch on the
 187 same grids. At the fastest speed the ensemble mean never reaches the fit
 188 window; 0.964 is a local log-slope of a delayed trajectory.

189 This delayed passage through a noisy bifurcation (Berglund & Gentz,

2002) makes the apparent exponent depend on ramp speed and fitting window. A deviation from the square-root law tests the adiabatic approximation where branch tracking is independently justified. The sweep does not bracket the specified deviation-based critical ramp speed.

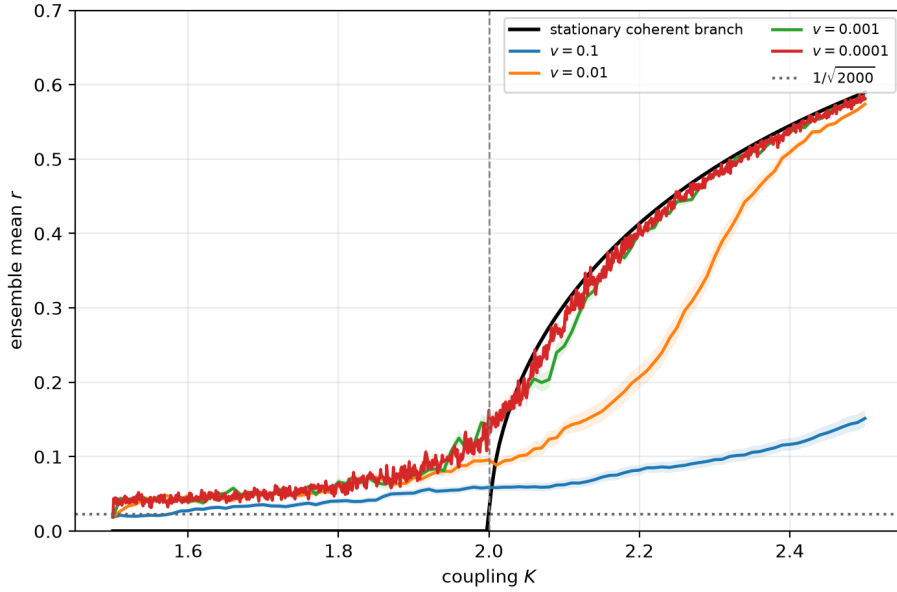


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ contains 2. These are finite numerical regimes, with full parameters and controls in Supplemental Material; they supply no cortical calibration.

199 **3.3 What the estimators produce on EEG**

200 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
201 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
202 tages produce different residuals. The bipolar results are compatible with the
203 Bessel relation, but their concentration range cannot distinguish it from its
204 linear approximation at the observed scatter. Temporal pooling changes the
205 estimand, and sedation recordings leave the awakening hypothesis untested.

206 An informative dataset needs broader concentration, calibrated estima-
207 tors and an observation model for dependence, source mixing and reference
208 choice. Public mouse calcium imaging provides thousands of spatially lo-
209 cated neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but
210 its 7.65 Hz frame rate is below the phase-estimation band used here. Site
211 count, state coverage and bandwidth are separate requirements.

212 **3.4 A protocol for the awakening experiment**

213 The physical test has six requirements:

- 214 1. **Coupling and diffusion in the same units.** Estimate them from re-
215 solved phase dynamics, separating heterogeneity and observation noise.
216 Measure the drive or perturb the system to distinguish endogenous cou-
217 pling.
- 218 2. **Geometry calibrated to coupling.** Validate the geometry observ-
219 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Adequate phase measurements.** The proposed spatial estimator uses at least 100 sites, resultant length for r , and unweighted regression of log density on centered cosine for a . At that site count its 40 bins clear the bias floor at no scanned concentration and no dependence, and 24 bins or fewer clear it only from $a = 1.5$ upward with independent sites. The observed ceiling is $a = 0.542$, so discriminating within the observed range takes 3000 independent sites instead, rising to 31000 when sites share a mean in groups of 10.

4. **An independent recovery endpoint.** Measure behavioral recovery separately from phase organization.

5. **A predeclared onset criterion.** Set a coherence threshold above the finite-size floor and account for ramp speed.

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A

240 favorable outcome still leaves the content of coordinated activity to be ex-
241 plained.

242 4 Compatibility: from coordinated activity 243 to a global description

244 4.1 Why common timing is insufficient

245 For overlapping patches U and V , compatibility means that restricting either
246 local description to $U \cap V$ gives the same object. Descriptions can agree on
247 shared quantities while contributing different information elsewhere. For the
248 spatial measures used here, compatible restrictions give a unique measure on
249 the covered domain: the gluing property formalized by a sheaf.

250 In the cup example, visual and reaching descriptions must express a
251 shared quantity, such as cup location, in common coordinates. Restriction
252 retains that quantity while dropping details particular to each description.
253 Agreement permits surface appearance and hand configuration to contribute
254 to one situation. The mathematical counterexample assigns the same phase
255 to every patch while two local mass profiles disagree on a shared site. Those
256 profiles have no common extension. Because they are chosen independently
257 of phase, synchrony places no constraint on their disagreement.

258 This distinction limits the biological interpretation. Receptive territories
259 and content domains must be related to spatial patches, and the descriptions

260 must support the required extension. Matching overlap marginals of de-
 261 coded probability laws need not determine a unique global joint distribution
 262 (Abramsky & Brandenburger, 2011). The spatial-measure result supplies a
 263 precise construction; applying it to perceptual content requires a justified
 264 content model.

265 4.2 When coherence does constrain content

266 A declared relation between phase and content gives a positive result. Let
 267 contents take values in a real normed vector space E . Patch i reads an
 268 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 269 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 270 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 271 sites; their equality on overlaps is a supplied relation between descriptions of
 272 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 273 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 274 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (8)$$

275 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 276 agree exactly. The factor N makes this estimate potentially weak in large
 277 populations; it supplies a pointwise guarantee from a population average. A
 278 shared scalar encoder is a special case. Dropping the overlap relation permits

disagreement even at perfect locking. A common nonexpansive linear projection preserves agreement, but agreement of a projection does not establish agreement of the full vector.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i + \eta o_i$. Content and observation discrepancies bounded by ε and δ give a next discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (9)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, reduce an initial mismatch. Approximate observations leave a nondecaying term in the error bound. Taking a uniform bound on Eq. (8) connects this update to the phase process. These are mechanisms for specified vector contents in shared coordinates; their neural meaning and relevance to experience remain to be established.

4.3 What approximate agreement determines

Exact gluing requires exact restrictions. For nonnegative profiles on a finite substrate, suppose every overlap discrepancy is at most ε in uniform distance. Weights that vanish outside their patches and sum to one at each site select a global profile by weighted averaging. The selection lies within ε of each patch there, and any two such selections differ by at most ε , independently of cover multiplicity. An L -Lipschitz readout consequently varies by at most

299 $L\varepsilon$ across selections.

300 At zero tolerance, the selection is the unique exact extension. At every
301 positive tolerance there are families with no exact extension; selection then
302 gives an approximation whose dependence on weights is bounded. Supple-
303 mental Material gives a three-site witness attaining the bound. The mass-
304 profile construction uses the scalar case of the content dynamics and addi-
305 tionally requires nonnegativity of the initial profiles and observations. The
306 general vector result supplies no identification with the spatial-measure sheaf.
307 Neither construction selects the neural cover or decoder.

308 4.4 Agreement after a change takes communication

309 Snapshot compatibility does not specify how quickly descriptions respond
310 when a shared quantity changes. In a synchronous network whose updates
311 read only the site’s own state and declared neighbours’ messages, two initial
312 configurations agreeing on a site’s causal past through T rounds produce the
313 same state there at round T . If two fresh interventions outside that past
314 require reports separated by more than 2ε , no report from that site can be
315 within ε of both. This follows from the reconstruction bound in Section 5.2.

316 A three-site delay line witnesses failure after one round and exact report-
317 ing after two; cutting the path prevents delivery at every round. A report
318 may still happen to be correct about one fixed world. The bound concerns
319 responsiveness across interventions. Applying it requires a justified commu-
320 nication graph, observation paths and deadline.

321 4.5 Coherent adaptation can lose environmental align- 322 ment

323 An adaptive coupling rule might supply representational structure, but its
324 objective must reward the relevant agreement. For symmetric coupling, pair-
325 wise sine terms cancel in the population mean drift. Minimizing squared drift
326 at that fixed mean therefore rewards equalizing drift across nodes. In the
327 frequency-cluster control, this favors coupling between unlike-frequency clus-
328 ters, whereas the diagnostic environmental template rewards coupling within
329 clusters.

330 For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (10)$$

331 The formalization constructs the drift operator A and adjoint, obtains $\nabla\sigma =$
332 $(2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This
333 functional differs from thermodynamic entropy production: at a coherent sta-
334 tionary density the probability current vanishes while phase-averaged squared
335 drift is positive.

336 The finite control co-evolves noisy phases and symmetric, nonnegative
337 couplings at fixed total resource. Its template is a diagnostic covariance of
338 latent signals, never a training input. Across three seeds, second-half order
339 remains between 0.9167 and 0.9222, while the within-cluster to between-
340 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).

341 Random-update, relabelled-gradient and frozen-kernel controls are reported
342 in Supplemental Material.

343 The immediate post-update objective reduction disappears when aver-
344 aged over complete update intervals. A prespecified grid of 6 rate-cadence
345 combinations contains no case meeting the 5% sustained-reduction criterion
346 while retaining coherence and losing alignment. The finding is coherent adap-
347 tation away from the specified structure, with no demonstrated sustained cost
348 reduction.

349 Kernel alignment and overlap compatibility are different observables.
350 This control addresses acquisition of its designated template under its par-
351 ticular objective. Learning models can explicitly include data in an objective
352 (Corberi et al., 2026); an account of agreement about the cup must specify
353 the content its learning rule is meant to preserve.

354 4.6 What must be specified and measured

355 A neural test needs independently identified domains, local descriptions and
356 restriction maps. Two decoders should estimate a shared content on over-
357 lapping domains before their agreement is examined. Discrepancy requires
358 a null under decoding noise and controls using deliberately compatible and
359 incompatible examples. Shared priors can manufacture agreement, while
360 regional bias can manufacture disagreement. A phase-agreement estimator
361 would measure the variable whose explanatory limits are at issue: on con-
362 structed content the phase carries a share 1.000 of the content entropy when

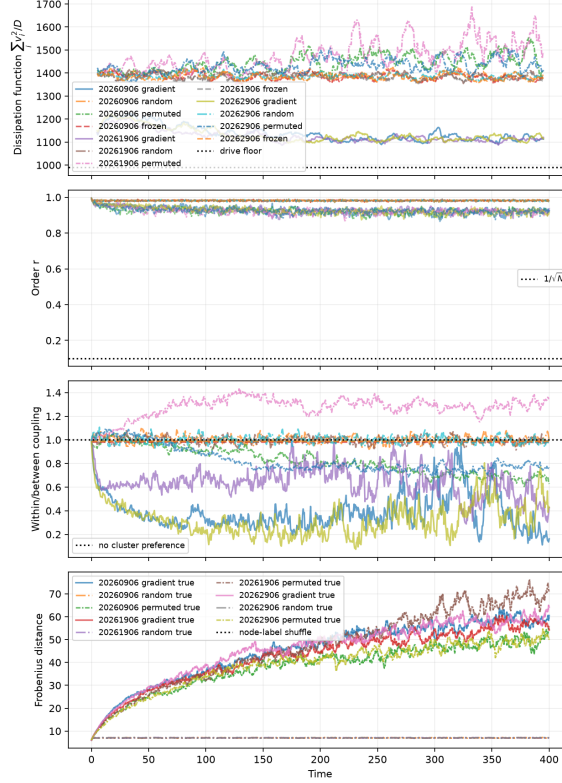


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

the content is built from it and 0.000 when it is not, and a phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. Supplemental Material reports an implementation meeting these requirements on constructed data.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The content mechanisms above identify additional hypotheses under which agreement can arise; they leave the empirical choice of content and the connection to that cover explicit.

5 Self-representation: a state that models itself

5.1 The reflexive construction

Compatibility gives a global state. Reflexivity relates it to an encoding within that state. Equip global sections with a complete metric. If the additional restriction-resonance condition identifies encoding with restriction to a region U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (11)$$

381 The region and readout are fixed across the state space. If F contracts and
 382 fixes the section s_* formed from the local descriptions, Banach's theorem
 383 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 384 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 385 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 386 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 387 obligation.

388 In the cup example, the global state includes the represented scene and the
 389 perceiver's relation to it. A local encoding must retain enough information
 390 for the readout to recover that modelled state; cup position alone would be
 391 insufficient if hand configuration and viewpoint could not be reconstructed.
 392 Here *prediction* names a reconstruction map. Interpreting it as temporal
 393 neural prediction requires dynamics.

394 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 395 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (12)$$

396 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 397 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 398 weakens. A biological interpretation must specify the represented variables,
 399 metric and acceptable fidelity; changing readouts need a further model.

400 **5.2 Reconstruction across a family of states**

401 Sensitivity can be required quantitatively. Declare relevant macrostates, an
402 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
403 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (13)$$

404 Thus M relevant states pairwise more than 2ε apart require at least M
405 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
406 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
407 existing self-prediction map.

408 This is an additional accuracy criterion on a family, with no requirement
409 for multiple exact fixed points. A blind encoder or constant readout fails it
410 for separated states despite having a unique fixed point. The family, metric
411 and tolerance are inputs; interpreting the code count as physical memory
412 capacity requires a hardware model.

413 **5.3 Why connect these properties to unity and a min-** 414 **imal self?**

415 We propose the state glued from compatible local descriptions as a correlate
416 of conscious unity, and its accurate internal reconstruction as a correlate of
417 a minimal self. The motivations are that different contents belong to one ex-

418 perience situation, and that the situation is experienced from a perspective.

419 Compatibility expresses agreement about shared quantities while allow-
420 ing different contributions. It also permits a description of uncertainty or
421 conflicting judgments: apparent cup position and judged cup position can
422 be represented as distinct quantities within a consistent account of their dis-
423 crepancy. Defining which quantities are shared is substantive. A choice made
424 solely to secure agreement would deprive the proposal of empirical force.

425 The minimal-self proposal concerns the perceiver’s position and condition
426 in that situation: where the cup appears relative to a body that can reach it.
427 Such an encoding could be implicit and need carry no autobiographical nar-
428 rative. Its proposed necessity is philosophical: even an unreflective episode
429 has a perspective. The gluing theorem alone requires no self-encoding.

430 Reconstruction also needs a functional interpretation. A reconstructable
431 recording need never be used to distinguish changes in a system from changes
432 in its surroundings. Altering the encoded hand or viewpoint while holding
433 the external scene fixed should change estimates of the perceiver’s relation
434 to the cup as the readout predicts. Whether that represented perspective is
435 experienced is a further question.

436 **5.4 Selecting and testing the construction**

437 Both uniqueness results concern specified objects. They choose neither the
438 cover nor the encoding, and do not select between candidate subjects. Inte-
439 grated information theory treats exclusion explicitly (Tononi et al., 2016); the

440 present construction leaves selection to independent evidence about neural
441 organization.

442 A constant readout returning a predetermined state contracts and has a
443 fixed point while discarding its encoding. It fails the functional sensitivity
444 required above. Local descriptions must predict the contents assigned to
445 their domains; encoding and readout must be specified before testing their
446 responses over a declared range.

447 A test of the full proposal must separately measure coupling relative to
448 diffusion, content compatibility and self-reconstruction, alongside an inde-
449 pendent endpoint for experience. Meeting all three without a unified episode
450 would challenge sufficiency. A unified episode with incompatible descrip-
451 tions would challenge the unity identification; one lacking the specified self-
452 encoding would challenge the reflexive requirement. Awakening, dreaming
453 and dissociative states offer possible settings, but absence of report cannot
454 by itself define absence of experience.

455 The compatibility observable is a specification exercised on constructed
456 data — it separates built-in incompatibility from built-in compatibility at
457 $AUC = 0.990$, and its two silent failures are caught by per-patch diagnostics
458 rather than by the statistic — and it has no neural instantiation, as the neural
459 self-encoding has yet to be identified. Gluing and fixed-point conditions
460 permit multiple substrates; the electromagnetic commitment enters through
461 physical calibration. The proposal concerns which state is conscious and
462 leaves unexplained why those relations would be accompanied by experience.

463 6 Thermodynamics and composition

464 The phase and representation results constrain a candidate implementation
465 through different quantities: coupling, content disagreement and reconstruc-
466 tion error. Thermodynamics additionally constrains its physical operations.
467 Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on
468 specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker,
469 2000). A separate boundary result concerns a field reaching disconnected
470 vacuum components (Kibble, 1976). Their physical connections are inputs
471 to the composition below.

472 6.1 Feedback changes the information constraint

473 For passive memory X_t driven by an environmental signal S_t , the predictive
474 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (14)$$

475 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
476 et al., 2012). Zero retained information also satisfies it, so a task and a
477 learning rule require additional specification. Relating this bound to Lan-
478 dauer heat requires a common thermal scale and budget, and memory waste
479 bounded by erased entropy.

480 An acting agent changes the signal it will receive. Specify policy, world,

481 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{15}$$

482 The second conditional term vanishes in the passive model. With feedback,
483 the signed difference can be negative as action creates correlations with the
484 future. A causal reading requires an environmental state sufficient to describe
485 the relevant dynamics.

486 For a controlled substep, the active thermodynamic bound instead uses
487 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
488 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
489 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
490 2013). Every forward path of positive probability must have a positive-
491 probability reverse for this finite identity. A missing reverse gives infinite
492 divergence. The supplement develops the support conditions and composes
493 actuation with memory updates on their shared intermediate law.

494 **6.2 Learning and repeated operation need explicit re-** 495 **sources**

496 The finite examples distinguish selecting a feasible policy, relaxing a register
497 toward an encoded reward, and learning an unknown task from observa-
498 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

One observation channel is derived from a declared phase configuration: sample an oscillator uniformly and report its phase through a specified finite readout. Symbol probabilities are population fractions. An informative four-oscillator sensor and an uninformative control have the same $r^2 = 1/4$ at every parameter and action. Under the same update rule, their expected rewards are respectively $3/4 - (1/4)2^{-n}$ and $1/2$ after n updates. Their difference lies in the phase-world relation and readout, which coherence alone does not specify.

Attaching the learner's register to the declared actuator of Section 2.2 ties reward, installed kernel and work to that same executed law. In this exact witness, $U_n = 3/4 - (1/4)2^{-n}$ and mean installation work per update is $(1 + \log 3)2^{-n}/8$; cumulative work stays within $(1 + \log 3)/4$. The control keeps $U_n = 1/2$ and spends zero mean work. At the witness's $\kappa = 4$ and $D = 1$, the control remains on the exclusion line while one learning update raises the learner above it. Clearing that necessary energy condition establishes no evolving coherent field: phase dynamics, installation and sensing are not closed into a feedback trajectory.

For that model, common energy E , thermal scale θ , local detailed balance

520 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (16)$$

521 where expected cumulative work stays within the supplied allowance B at
 522 every prefix. This bounds expectations. Separate constructions track a store
 523 on each trajectory, include finite sources, charge sensor clearing and prepend
 524 preparation. They specify when a mean allowance represents an available
 525 resource; an expected budget alone permits overdrawn paths.

526 These constructions establish costs for their declared processes. A finite
 527 source with a positive net cost permits only finitely many funded cycles,
 528 while continuing operation requires replenishment. Register–bath witnesses
 529 also distinguish actual bath-energy transfer from transition-log heat. Ident-
 530 ifying a work store, an erasure budget and the installation energy of Sec-
 531 tion 2.2 requires physical relations that these separate models do not supply
 532 automatically.

533 **6.3 What the composition establishes**

534 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 535 cept eight connecting implications (Table 1). In the active branch, a named
 536 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (17)$$

537 with B the register’s heat. The upper allocation follows from a ledger whose
538 operations deliver that total heat to one reservoir and whose other operations
539 do not increase joint entropy. Those ledger assumptions are physical inputs.
540 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (11).

Joint witnesses combine a register, boundary, refining mesh and small spatial cover; active witnesses add controlled thermal updates and a priced installation. They establish satisfiability under their stated identifications. Mesh convergence follows from refinement data, and the witnesses establish no common cortical mechanism. Supplemental Material maps the claims to their Lean declarations and records the counterexamples. The development

553 declares no additional axioms beyond its libraries' foundational axioms.

554 **7 Discussion**

555 The framework distinguishes the physical conditions for coherent activity
556 from the relations needed to interpret it as one internally represented sit-
557 uation. Its positive connections depend on specified mechanisms: installed
558 modes relate energy to mean coupling, phase encoders shared on overlaps
559 relate coherence to vector-content agreement, and an observation update
560 propagates that agreement into retained fields. Their assumptions identify
561 what a cortical model must establish.

562 The separation results delimit what would count as evidence. A station-
563 ary recovery distribution leaves coupling, diffusion and a matched common
564 drive indistinguishable. Synchronous patches can carry incompatible phase-
565 independent descriptions. Coherent adaptation under the tested squared-
566 drift objective can lose environmental alignment. Each result concerns the
567 stated model and observable; none determines cortical learning from ther-
568 modynamics or diagnoses experience from synchrony alone.

569 For the awakening hypothesis, the next requirement is a calibrated
570 geometry-coupling relation and resolved phase dynamics, with an indepen-
571 dent recovery endpoint. The installation bound adds a necessary resource
572 condition where its hardware representation applies. It does not calibrate
573 cortical mode prices or replace evidence that the physical field operates

574 above threshold.

575 For the representational connection, the informative experiment is
576 whether a controlled change in coordination alters independently decoded
577 content agreement. Decoders and restriction maps must first recover compat-
578 ible and incompatible cases under realistic noise. Perturbations then need
579 controls for sensory information and decoder quality. The phase-encoder
580 theorem supplies one mechanism to test, but shared overlap encoders, com-
581 mon coordinates and population-size dependence limit how directly it can
582 describe neural content. Equal coherence can also accompany different task
583 information, as the phase-sensor control demonstrates.

584 The self clause requires a candidate encoding whose perturbation changes
585 the represented scene, body or viewpoint as its readout predicts. Approx-
586 imate gluing and reconstruction make fidelity an explicit parameter. They
587 leave the cover, metric, encoder and required resolution to be justified inde-
588 pendently; uniqueness within these choices does not choose a subject.

589 Integrated-information, global-workspace and predictive-processing ac-
590 counts organize unity, access and modelling differently (Tononi et al., 2016;
591 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal con-
592 tributes a candidate physical interaction and conditional relations between
593 local content and self-encoding. Its explanatory value depends on identifying
594 those relations in neural activity and showing that they matter for experi-
595 ence. In the cup example, the challenge is to explain how coordination helps
596 maintain one represented situation and how the perceiver’s place within it

597 contributes to the episode.

598 **Data and code availability**

599 The Lean development, simulation scripts, tests and saved numerical sum-
600 maries are maintained in the manuscript source repository. The numerical
601 routines carry property-based tests beside their fixed-case tests, checking in-
602 variants over generated inputs: conservation of total coupling, symmetry and
603 zero diagonal under the resource projection and the plasticity gradient, the
604 zero and coherent branches of the stationary order parameter either side of
605 $K_c = 2D$, row balance and excitatory–inhibitory sign structure of the frus-
606 trated network, normalization and reflection symmetry of the spatial kernel,
607 and agreement of each reduced mean-field evaluation with its explicit pair
608 sum. Publication simulation values are regenerated from saved summaries
609 without rerunning production sweeps. The exploratory EEG analysis uses
610 OpenNeuro dataset ds005620 (Bajwa et al., 2025); preprocessing and selec-
611 tion are described in Supplemental Material. The source repository contains
612 the reproducible analysis and does not redistribute the raw recordings.

613 **Funding and competing interests**

614 This research received no external funding. The author declares no compet-
615 ing interests.

616 Use of AI tools

617 AI assistance to this work was substantial and is not separable by task.
618 Several coding assistants and chat interfaces were used in an interleaved way
619 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
620 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code
621 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
622 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
623 they contributed to the Lean development, the simulation and analysis code,
624 and the drafting and revision of both documents. Because the tools were
625 used in combination and in alternation, no section, proof, figure or result is
626 attributable to any one of them, and none is claimed to be. The author is
627 responsible for all content, including everything a tool produced. Machine
628 checking establishes the stated Lean results under their hypotheses; it does
629 not validate biological interpretations or replace scientific review of generated
630 code and prose.

631 References

- 632 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
633 electric fields emerge from and allow representational drift. *NeuroImage*,
634 253, 119058.
- 635 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-

tion theory: from consciousness to its physical substrate. *Nature Reviews*
Neuroscience, 17(7), 450-461.

Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
 Mind. *Oxford University Press*.

Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-
 tational theory of conscious processing. *Current Opinion in Neurobiology*,
 25, 76-84.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
Reviews Neuroscience, 11(2), 127-138.

Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
 40, 24-50.

Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
 parts of the free energy principle. *Entropy*, 23(3), 293.

Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:

655 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
656 *Progress in Neurobiology*, 226, 102465.

657 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
658 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
659 9(8), 45-60.

660 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
661 testable hypothesis about the characteristics of conscious as opposed to
662 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

663 McFadden, J. (2020). Integrating information in the brain's EM field: the
664 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
665 niaa016.

666 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
667 guide neocortical network activity. *Neuron*, 67(1), 129-143.

668 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
669 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
670 95-103.

671 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
672 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

673 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
674 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

675 Verkhatsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
676 *logical Reviews*, 98(1), 239-389.

677 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
678 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
679 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

680 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
681 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
682 ity of complexity measures to capture dreaming during propofol sedation.
683 *Scientific Reports*, 15, 32746.

684 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
685 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
686 brain. *Science*, 342(6156), 373-377.

687 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
688 *Physiological Reviews*, 88(4), 1277-1340.

689 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
690 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

691 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
692 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
693 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

694 Landauer, R. (1961). Irreversibility and heat generation in the computing
695 process. *IBM Journal of Research and Development*, 5(3), 183-191.

696 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
697 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

698 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
699 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

700 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
701 of prediction. *Physical Review Letters*, 109(12), 120604.

702 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
703 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

704 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
705 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
706 031015.

707 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
708 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
709 PhysRevLett.111.180603.

710 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
711 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

712 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
713 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
714 181.

- 715 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
716 archive.pitt.edu/115.
- 717 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
718 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
719 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 720 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
721 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
722 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 723 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
724 bifurcations with additive noise. *Probability Theory and Related Fields*,
725 122, 341–388. doi:10.1007/s004400100174.
- 726 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
727 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
728 doi:10.1088/1367-2630/13/11/113036.
- 729 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
730 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
731 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
732 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 733 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
734 terns underlie complex brain dynamics and are related to cognitive pro-

cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
 023-01626-5.

Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
 and dissipation with partially observed fluctuating signals. *Physical Review
 Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
 cell resolution functional networks during unconsciousness are segre-
 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
 doi:10.1016/j.celrep.2025.116902.

Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.